
November 28, 2025

Assignment problems (10%)

1. Solve problem #7 from chapter one end exercises.
2. Solve problem #19 from chapter one end exercises.
3. Consider a track vehicle moving at a speed of 10 m/s at $t = 0$. If the vehicle accelerates $a = 3\text{m/s}^2$ until $t = 3\text{s}$, determine the total distance travelled between $t = 0$ and $t = 3\text{s}$, using Trapezoid and Simpson methods. Do this in $n = 1, 3, 5, 10$ slices and tabulate the values. Hint: Use the integration function

$$S = \int_0^3 (v_0 + at)dt,$$

as the total distance travelled.

4. A particle is thrown vertically up into the air from a top of a building with initial velocity of $v_0 = 10\text{m/s}$. After what time t would the particle strike-back the building? How much displacement S has the particle travelled during this time interval?
Hint: [Use one of the root finders to solve for t & one of the integration solvers to solve for S . $g = 10\text{m/s}^2$].
5. Consider a differential equation

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0.$$

Determine the solution using either of the initial valued problem solvers if $x(0)=0$, and $x'(0)=1$. Plot $x(t)$ vs t .

6. In question #5, if the problem represents an oscillation of a bond in O_2 molecule, how does the total energy E vs t change by applying an LJ potential and using molecular dynamics within **Euler** or **Verlet** algorithms and a **monte carlo** scheme?